HYDERABAD PROPERTY PRICE PREDICTION PLATFORM

PropGPT — Comprehensive Implementation Plan

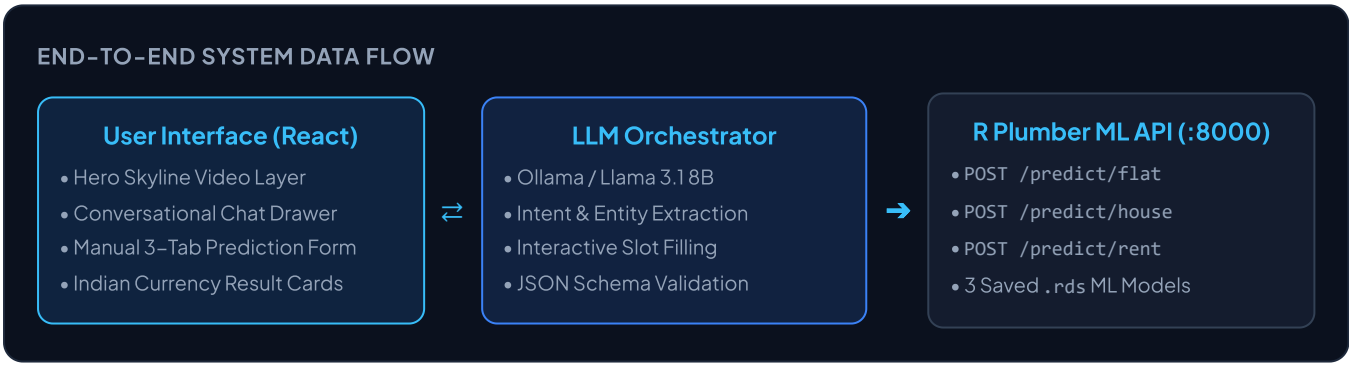
Architectural blueprint and technical specification for building a production-quality, responsive React frontend integrated with an LLM conversational orchestrator and three independent R Plumber machine learning prediction models.

Focus City: **Hyderabad, Telangana, India** Frontend: **React 18 + Vite + Tailwind CSS**

ML Engine: **R Plumber (Port :8000)** LLM Backend: **Ollama (Llama 3.1 8B)** Version: **1.0.0**

1 Executive Summary & System Architecture

PropGPT combines a cinematic real-estate aesthetic with conversational AI and trained statistical/machine learning models. Users can interact via two complementary paths: a *ChatGPT-inspired conversational interface* with slot-filling extraction, and a *structured manual prediction form* for deterministic property valuation.



> **Architectural Boundary Guarantee:** The React frontend does not call the R Plumber prediction endpoints directly from the LLM conversation flow. The orchestration backend strictly validates the extracted JSON payload before invoking Plumber, preventing invalid requests or hallucinated prediction figures. The manual form calls Plumber through a centralized API service layer.

2 Visual Identity & Design System

The visual direction follows a modern, dark, high-trust real estate technology aesthetic, inspired by Financial District and HITEC City skylines. Exaggerated glowing effects, floating AI particles, and generic AI templates are avoided in favor of restrained glassmorphism and clean typography.

Design Token	Color / Value	Usage & Implementation Details
Primary Background	`#0B111E` (Dark Navy)	Base viewport color, footer background, and deep container base.
Surface / Card	`#141C2E` (Charcoal Navy)	Chat panel, manual form cards, and result containers.
Accent Colors	`#38BDF8` (Sky Cyan) & `#2563EB` (Cobalt)	Interactive buttons, active tab indicators, icons, and keyword highlights.
Typography	Inter / Plus Jakarta Sans	High-contrast, clean legibility across numeric property specs and text.
Glassmorphism	`backdrop-blur-md bg-[#141C2E]/70 border-white/10`	Subtle translucent elevation for the chat box, navbar, and cards.
Skyline Video	`public/videos/hyderabad-skyline.mp4`	Full-screen muted loop with dark navy gradient overlay (contrast guarantee).

3 Project Directory & Component Hierarchy

The frontend codebase is organized into modular components, services, and utility helpers:

```
PropGPT/
├── frontend/
│   ├── public/
│   │   ├── videos/hyderabad-skyline.mp4       # Hyderabad skyline video loop
│   │   ├── images/skyline-poster.webp         # Video loading fallback poster
│   │   └── favicon.ico
│   ├── src/
│   │   ├── components/
│   │   │   ├── common/Navbar.jsx              # Frosted navbar with brand & anchors
│   │   │   ├── common/Footer.jsx              # Tech stack attribution & disclaimer
│   │   │   ├── common/FormControls.jsx        # Custom Select, Slider, Input, Tooltip
│   │   │   ├── hero/HeroSection.jsx           # Location indicator, title, subtitle
│   │   │   ├── hero/BackgroundVideo.jsx       # Viewport video with navy overlay
│   │   │   ├── hero/SuggestionCards.jsx       # 4 prompt starter cards
│   │   │   ├── chat/ChatInput.jsx             # Auto-resizing textarea, mic, send
│   │   │   ├── chat/ChatInterface.jsx         # Conversation drawer & message stream
│   │   │   ├── chat/MessageBubble.jsx         # User & PropGPT message wrappers
│   │   │   ├── chat/PredictionCard.jsx        # Structured result display (INR formatted)
│   │   │   ├── manual-form/ManualPredictionForm.jsx # Segmented 3-tab container
│   │   │   ├── manual-form/FlatForm.jsx       # Flat Purchase inputs (Model 1)
│   │   │   ├── manual-form/HouseForm.jsx      # House Purchase inputs (Model 2)
│   │   │   ├── manual-form/RentForm.jsx       # Monthly Rental inputs (Model 3)
│   │   │   ├── sections/HowItWorks.jsx        # 4-step workflow cards
│   │   │   ├── sections/WhyPropGPT.jsx        # Value propositions & credibility
│   │   │   └── sections/AboutSection.jsx      # Methodology & ML background
│   │   ├── services/
│   │   │   ├── plumberApi.js                  # Axios client for R Plumber (:8000)
│   │   │   └── chatApi.js                     # Ollama / Orchestration backend API
│   │   ├── utils/
│   │   │   ├── formatters.js                  # INR Lakh/Crore and rate/sq.ft helpers
│   │   │   └── validators.js                  # Form validation schemas
│   │   ├── constants/localities.js            # Hyderabad localities list
│   │   ├── App.jsx / main.jsx / index.css     # Master layout & Tailwind tokens
│   ├── .env.example                           # VITE_PLUMBER_API_URL=http://127.0.0.1:8000
│   └── package.json
```

4 ML Models & Dynamic Manual Form Specifications

The platform serves three independent trained R machine learning models. The manual form features a 3-tab switcher and conditionally renders only the fields relevant to each model:

Attribute / Field	Field Type & Permitted Values	Model 1: Flat	Model 2: House	Model 3: Rent
Locality	Dropdown (Kokapet, Gachibowli, HITEC City, etc.)	Required	Required	Required
Property Area (sq. ft.)	Number input (> 100 sq.ft.)	Required	Required	Required
BHK	Select / Stepper (1, 2, 3, 4, 5+)	Required	Required	Required
Bathrooms & Balconies	Numeric Steppers (Baths: 1–6, Balconies: 0–4)	Required	Required	Required
Furnishing	Select (Furnished, Semi-Furnished, Unfurnished)	Required	Required	Required
Area Source	Select (Super Built-up, Carpet Area, Built-up)	Required	Required	Required
Transaction Type	Select (New Booking, Resale)	Required	Required	Not Used
Property Floor & Total Floors	Number inputs (Property Floor ≤ Total Floors)	Not Used	Required	Required
Ownership	Select (Freehold, Leasehold, Co-operative)	Not Used	Required	Not Used
Facing	Select (East, West, North, South, North-East, etc.)	Not Used	Required	Not Used

5 Conversational AI & Backend API Integration

Conversational AI Flow

1. **User Query:** User types natural-language inquiry (e.g. *"I want to buy a 3 BHK flat in Kokapet around 1800 sq ft"*) or clicks a suggested prompt card.
2. **Intent Extraction:** LLM orchestrator recognizes `FLAT_PURCHASE` intent and extracts slots: `{ locality: "Kokapet", bhk: 3, area: 1800 }`.
3. **Slot Validation & Follow-up:** If required fields are missing (e.g. `furnishing`), the LLM queries the user interactively.
4. **Execution:** Once all required attributes are gathered, the orchestrator issues a POST request to R Plumber.
5. **Structured Response:** The prediction returned by Plumber is rendered using the custom `PredictionCard` inside the chat stream.

R Plumber API Contracts

Endpoint	Method	Payload Schema (JSON)	Sample Response
`/predict/flat`	POST	`{ area, bhk, locality, furnishing, transaction_type, area_source, bathrooms, balconies }`	`{ "predicted_price": 18500000, "status": "success" }`
`/predict/house`	POST	`{ area, bhk, locality, furnishing, transaction_type, area_source, bathrooms, balconies, ownership, facing, property_floor, total_floors }`	`{ "predicted_price": 34000000, "status": "success" }`
`/predict/rent`	POST	`{ area, bhk, locality, furnishing, area_source, bathrooms, balconies, property_floor, total_floors }`	`{ "predicted_rent": 42000, "status": "success" }`
`/health`	GET	None	`{ "status": "healthy", "models_loaded": 3 }`

Prediction Result Card Specification

Every prediction result (whether via Chat or Manual Form) is presented with high-trust financial formatting:

- **Indian Numbering Format:** Values $\ge 1\text{ Crore}$ are formatted as `₹ X.XX Cr`; values $\ge 1\text{ Lakh}$ as `₹ XX.X Lakh`; rent as `₹ XX,XXX / month`.
- **Calculated Price Per Sq.Ft:** Calculated deterministically as `price / area` (e.g., `₹ 10,277 / sq.ft`).
- **Property Specifications Pill Bar:** Badges for Locality, BHK, Area, Floor, and Furnishing status.

> **Mandatory Statutory Disclaimer:**
> "PropGPT provides machine learning-based estimates for informational purposes. These estimates are not certified property valuations or guaranteed market prices."

6 Phased Implementation Roadmap

Phase 1: Foundation & Design System — SPRINT 1

- Initialize React + Vite + Tailwind CSS project
- Configure color tokens, glassmorphism, fonts
- Integrate Hyderabad skyline video loop & poster
- Build responsive Navbar with brand logo

Phase 2: Hero & Conversational Panel — SPRINT 1–2

- Implement Hero typography and location badge
- Build auto-resizing ChatInput with action buttons
- Add 4 clickable suggestion prompt cards
- Implement ChatInterface drawer and state transitions

Phase 3: Structured Manual Form — SPRINT 2

- Create 3-tab switcher (Flat, House, Rent)
- Dynamic field matrix with client-side validation
- Build centralized Plumber API service client
- Design INR currency formatted PredictionCard

Phase 4: LLM Orchestrator Bridge — SPRINT 3

- Connect chat interface to Ollama / Llama 3.1 8B
- Implement structured JSON slot-filling prompts
- Embed interactive prediction cards in chat stream
- Handle error states, timeouts, and network retries

Phase 5: Supporting Sections, Cross-Device Polish & Acceptance Testing — SPRINT 3–4

- Build *How It Works* (4-step cards), *Why PropGPT*, *About Section*, and *Footer*
- Perform mobile, tablet, and ultra-wide layout audit to ensure zero visual degradation
- Ensure keyboard accessibility (tab navigation, Enter-to-submit) and contrast compliance
- Verify end-to-end against live R Plumber API (:8000)